SQL Stored Procedures

A **stored procedure** is a group of SQL statements stored in a database that can be executed whenever required.

1. Basic Procedure

Example: Display all employee details.

```
DELIMITER //

CREATE PROCEDURE GetEmployees()
BEGIN
  SELECT * FROM Employee;
END //

DELIMITER ;

CALL GetEmployees();
```

Explanation:

- CREATE PROCEDURE – Creates a procedure.
- BEGIN and END – Define the procedure body.
- CALL – Executes the procedure.

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2. Procedure for INSERT

Example: Insert employee details into the table.

```
DELIMITER //

CREATE PROCEDURE InsertEmployee(
  IN id INT,
  IN name VARCHAR(50),
  IN salary INT
```

```
)
BEGIN
  INSERT INTO Employee
  VALUES (id, name, salary);
END //

DELIMITER ;

CALL InsertEmployee(1, 'Rahul', 25000);
```

Explanation:

- IN passes values to the procedure.
- INSERT INTO adds a new record.
- CALL executes the procedure with the given values.

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3. Procedure for UPDATE

Example: Update an employee's salary.

```
DELIMITER //

CREATE PROCEDURE UpdateEmployee(
  IN id INT,
  IN new_salary INT
)
BEGIN
  UPDATE Employee
  SET salary = new_salary
  WHERE emp_id = id;
END //
```

```
DELIMITER ;

CALL UpdateEmployee(1, 30000);
```

Explanation:

- UPDATE modifies existing records.
- SET specifies the new value.
- WHERE identifies the employee to update.

Note: Here, emp_id is the employee ID column.

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4. Procedure for DELETE

Example: Delete an employee using their ID.

```
DELIMITER //

CREATE PROCEDURE DeleteEmployee(
  IN id INT
)
BEGIN
  DELETE FROM Employee
  WHERE emp_id = id;
END //

DELIMITER ;

CALL DeleteEmployee(1);
```

Explanation:

- DELETE FROM removes a record.
- WHERE specifies which record to delete.
- CALL executes the procedure.